

Title

Laboratory time series moisture manipulative experiment from sediment across San Antonio, Texas: aerobic respiration and geochemistry

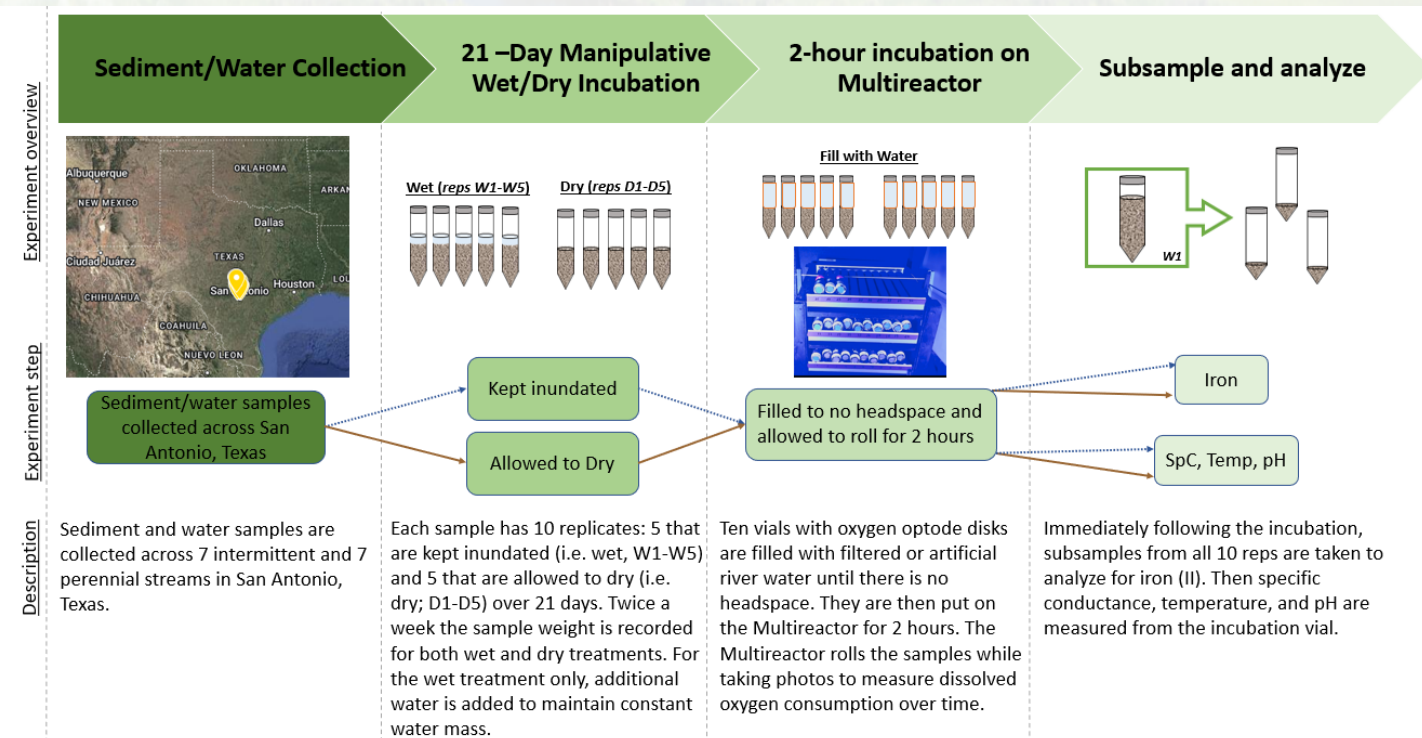
Summary

This dataset supports a broader study examining the effects of wetting and drying on hyporheic zone respiration. The dataset provides data generated from a laboratory moisture manipulation experiment. The contents include time series aerobic respiration and moisture; dissolved oxygen; sediment geochemistry data; and field metadata (including qualitative information on instream and river corridor characteristics).

Samples were collected as part of the WHONDRS Allison Veach collaboration (AV1). The data package associated with the AV1 study is available at <https://data.ess-dive.lbl.gov/datasets/doi:10.15485/2529428>. AV1 sampling occurred across 7 perennial and 7 intermittent streams in San Antonio, Texas. Each stream/site was visited both in summer during base flow (July-September 2023) and winter during peak flow (January-February 2024). This study uses subsamples from a subset of AV1 samples. The original field samples were labeled as AV1_###. Subsequent subsamples for this study were labeled as EV_###. The labels from the field samples and the EV subsamples can be mapped directly based on the digits following the prefix and underscore (i.e., EV_001 is a subsample from AV1_001). See the critical details section below for more details on sample naming.

For details on how to navigate data packages generated by this project, see <https://data.ess-dive.lbl.gov/portals/PNNLRiverCorridorSFA/About>.

Experimental Design

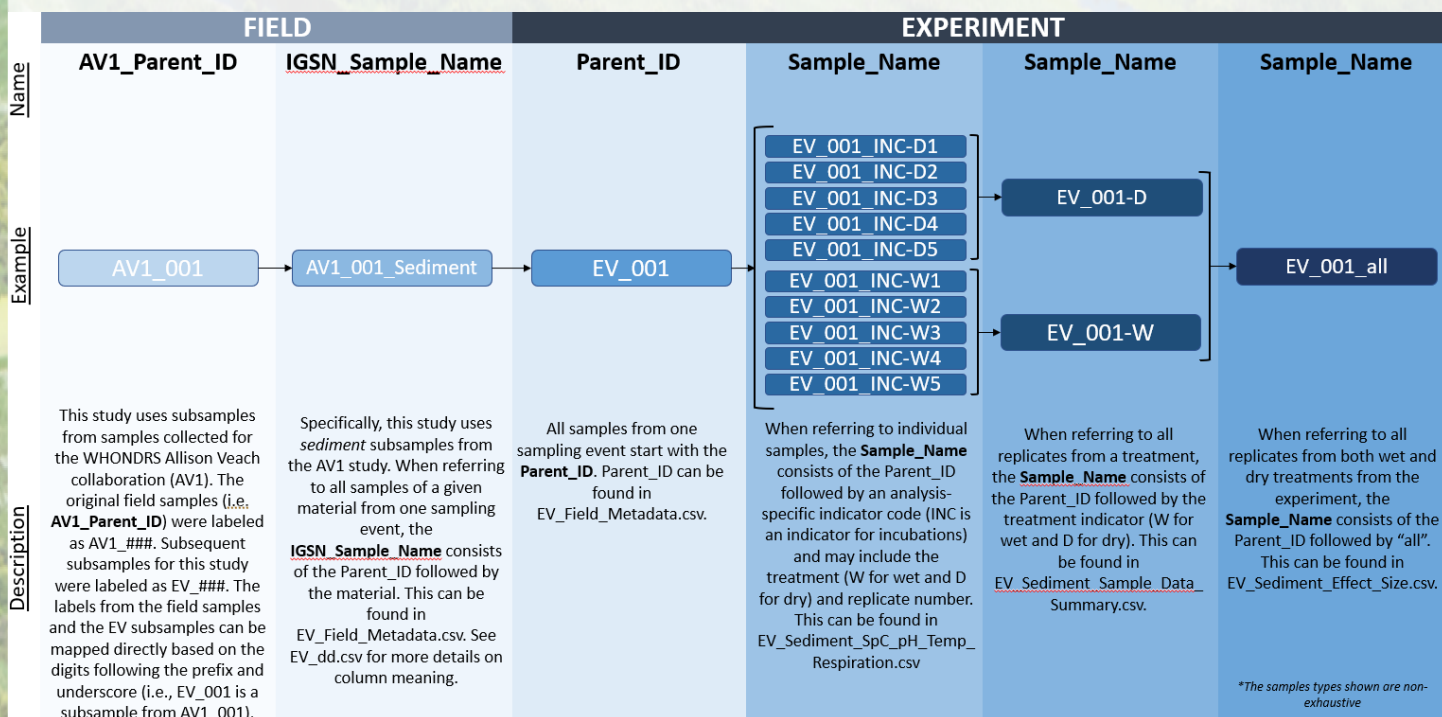


Brief Overview of Methods

Samples were collected in San Antonio, Texas via WHONDRS (<https://whondrs.pnnl.gov>) collaboration. Site metadata, including general environmental information, were manually recorded in the field. See WHONDRS_AV1_Sampling_Protocol.pdf for details on sample collection, sensor deployment, and site metadata collection. Briefly, if water was present, a miniDOT logger was deployed underwater upstream of the sampling site to measure dissolved oxygen and temperature during sampling. If no water was present, a miniDOT was not deployed. Unfiltered surface water and filtered surface water were collected at 50 percent water column depth. Sediment samples were collected underwater at 1 to 3 centimeters depth below the streambed and sieved to less than 2 millimeters. The samples were refrigerated after sampling and shipped or transported on blue ice to Pacific Northwest National Laboratory. Sediments were either kept constantly inundated or allowed to air-dry over 21 days while being shaken to maintain aerobic conditions. After 21 days, sediment aerobic respiration rate was calculated from time series dissolved oxygen data measured from laboratory sediment/water incubations using a non-invasive, multireactor method using customized oxygen optodes (Kaufman et al.; 2023). For details regarding laboratory and instrument methods for this analysis and others, see the alphanumeric methods codes located in the header rows of the sample data csv file and their associated definitions in EV_Methods_Codes.csv. Data formats follow ESS-DIVE Reporting Formats.

Critical Details

- 1 – Naming structure: Each sampling event has a unique Parent_ID in the format EV_###. The field metadata and data files all contain these unique IDs and can be mapped across each other accordingly. The Parent_ID may have other indicators appended when referring to samples. See figure below for details.



- 2 – Use the Site_ID in the Field Metadata to determine which unique sampling events (indicated with unique Parent_IDs in the Field Metadata) occurred at the same site at different points in time (i.e., baseflow and peak flow).

Data Package Structure

This dataset is comprised of one main data folder containing (1) file-level metadata; (2) data dictionary; (3) field metadata; (4) readme; (5) field protocol; and a (6) a subfolder with sediment sample data from the incubation experiment. The sample data subfolder contains (1) effect size; (2) iron (II); (3) gravimetric moisture; (4) respiration rates; (5) raw dissolved oxygen values and plots; (6) specific conductance; (7) pH; (8) temperature; (9) a summary containing mean, median, and standard deviation values of each data type for each treatment (wet and dry); and (10) methods codes. All files are .csv or.pdf.

Citations and Acknowledgements

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Citations:

- Kaufman MH, Torgeson J, Stegen JC (2023) Metabolic multireactor: Practical considerations for using simple oxygen sensing optodes for high-throughput batch reactor metabolism experiments. PLoS ONE 18(7): e0284256. <https://doi.org/10.1371/journal.pone.0284256>

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Change History

Change history:

Data Package Version	Changes
Version 1 <i>April 2025</i>	Original data package publication